**GROUP WORK PROJECT #** 2
**GROUP NUMBER:** 9284

MScFE 652: PORTFOLIO MANAGEMENT

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**Remember: Any group members who did **not** contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

| **Statement of integrity:** By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above). | |
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| Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.<br>**Note:** You may be required to provide proof of your outreach to non-contributing members upon request. |
|---|
| N/A |

# Step 1: Gathering Data

To construct the portfolio we considered data from 2021-01-04 to 2024-12-31. Then, to evaluate the performance of the portfolio we are going to use the data from 2025-01-06 to 2025-05-08. We downloaded the stocks for the following S&P 500 tickers from yahoo finance:

| Sectors/company | | | | | |
|---|---|---|---|---|---|
| Materials | LIN (Linde plc) | APD (Air Products and Chemicals Inc.) | ALB (Albemarle Corporation) | | |
| Communication Services | META (Meta Platforms Inc.) | CMCSA (Comcast Corporation) | FOXA (Fox Corporation) | GOOGL (Alphabet Inc.) | |
| Energy | XOM (Exxon Mobil) | COP (ConocoPhillips ) | APA (APA Corporation) | CVX (Chevron Corporation) | |
| Financials | JPM (JPMorgan Chase & Co) | PNC (PNC Financial Services) | LNC (Lincoln National) | BRK-B (Berkshire Hathaway) | V (Visa Inc.) |
| Industrials | CAT (Caterpillar Inc.) | UPS (United Parcel Service) | CAPR (Carrier Global Corporation) | GE (General Electric ) | |
| Information Technology | NVDA (NVIDIA Corporation) | SNPS (Synopsys Inc.) | ZS (Zscaler Inc.) | AAPL (Apple Inc.) | MSFT (Microsoft ) |
| Consumer Staples | PG (Procter & Gamble Co.) | KMB (Kimberly-Clark Corp.) | TAP (Molson Coors Beverage) | COST (Costco Wholesale Corp.) | |
| Utilities | NEE (NextEra Energy Inc.) | DUK (Duke Energy Corp.) | AES (AES Corp.) | | |
| Health Care | UNH (UnitedHealth Group) | MDT (Medtronic plc) | BAX (Baxter International) | LLY (Eli Lilly) | |
| Consumer Discretionary | AMZN (Amazon Inc) | NKE (NIKE Inc.) | ETSY (Etsy Inc.) | TSLA (Tesla Inc.) | |

| | | | | | |
|---|---|---|---|---|---|
| Real Estate | PLD (Prologis Inc.) | AMT (American Tower Corp) | AVB (AvalonBay Communities) | | |
| Extra Assets | GC=F (Gold) | BTC-USD (Bitcoin) | | | |

- We used the Close feature column and calculate the geometric returns as formula:

  rt=log(Pt/Pt−1)

  then we calculated mean returns.
- We downloaded daily 2-year Treasury yield from the FRED database (DGS2 series) as Risk Free Rate of return. Converted annual rate to daily risk free rate by dividing by trading days (as 252). Then we calculated the mean daily risk free rate.

### Normalized Stock prices growth Plots for Train and Test data:

**<Pls see pdf of ppt or jupiter>**

It can be seen that in the historical data used to construct the portfolio there are some assets that are outperforming the majority. But for the testing period we don't observe this discrepancy. In fact, the assets that are outperforming the others aren't the same as the training period.

To estimate the covariance of returns we used two approaches — one using the sample estimator, and one using the Ledoit–Wolf shrinkage estimator.

- The Sigma_sample is a matrix where diagonal elements are variance of each asset and off-diagonal represents covariance between pairs of assets.
- The Ledoit–Wolf shrinkage estimator is more stable and is recommended for practical applications, specially for large portfolios.

**Simga_sample and Sigma_shrinkage Estimator: <pls see pdf of ppt or jupiter>**

## Step 2: Markowitz Optimization

To determine the weights of the assets in the portfolio we employed the classical mean-variance model. In our portfolio we don't allow for shorting and also the maximum weight of an asset can be 0.35. In the optimization process we examined different upper bounds from 0.15 to 0.40. The performance of the model is highly affected by this threshold. Lower values of threshold helps in a better diversification but not necessarily better performance of the portfolio during the selected testing time.

The function implements mean-variance optimization as condition trying to find the maximum Sharpe ratio portfolio, assuming:

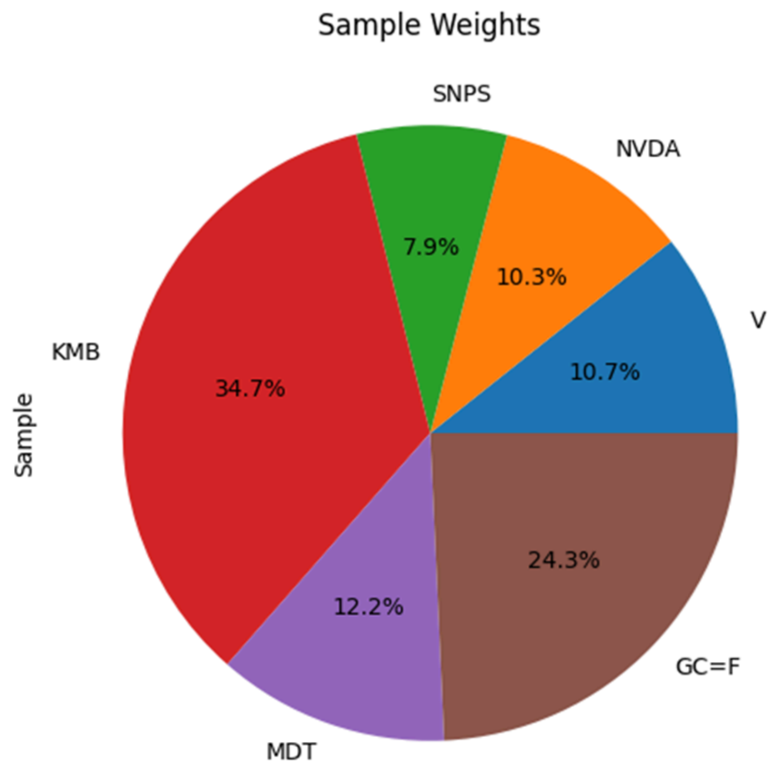
- A **long-only constraint** (no short selling),
- Optional **upper bounds** on asset weights,
- Use of the **SLSQP optimizer** from scipy.optimize.

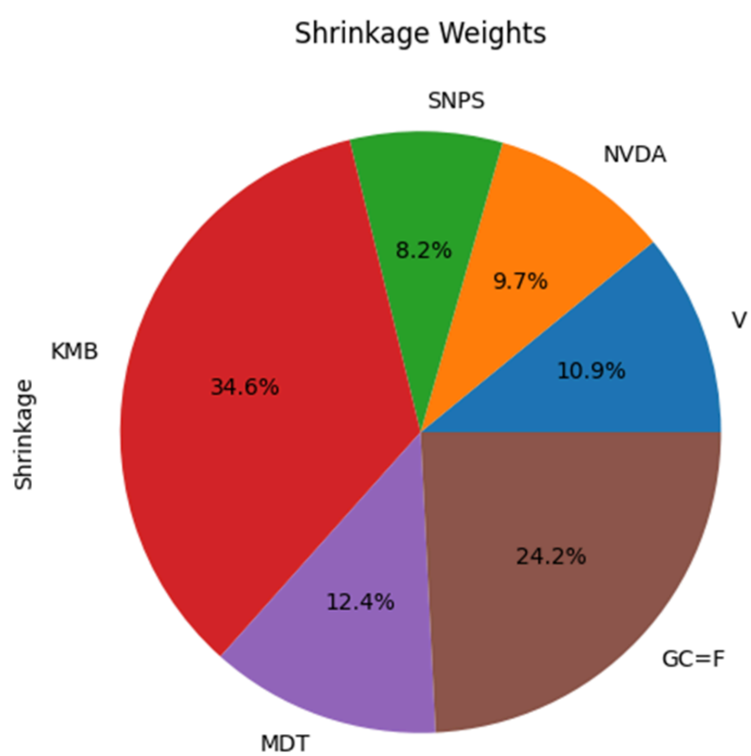
where

- w: Portfolio weights (vector)
- mu: Expected daily returns (vector)
- Σ: Covariance matrix of returns
- rf: Daily risk-free rate
- ·portf_mu = w @ mu - rf # daily excess return
- ·portf_sig = np.sqrt(w @ Σ @ w)
- ·portf_sharpe_ratio= port_mu / port_sig
- constraints:
  - bounds: Each asset weight is between 0 and w_max (default 1.0 → no leverage).
  - cons: A constraint that ensures the portfolio is fully invested (weights sum to 1).
  - w0: Starting point for optimization (equal weighting).

The resulting weights for two approaches of calculating the covariance are as follows. It is worthwhile to mention that, here we are displaying only the assets that were selected in the optimization process.

| | Sample | Shrinkage |
|---|---|---|
| V | 0.107 | 0.109 |
| NVDA | 0.103 | 0.097 |
| SNPS | 0.079 | 0.082 |
| KMB | 0.347 | 0.346 |
| MDT | 0.122 | 0.124 |
| GC=F | 0.243 | 0.242 |


Sample Weights
SNPS
NVDA
V
KMB
Sample
7.9%
10.3%
10.7%
34.7%
24.3%
12.2%
GC=F
MDT


Shrinkage Weights
SNPS
NVDA
V
KMB
Shrinkage
8.2%
9.7%
10.9%
34.6%
24.2%
12.4%
GC=F
MDT

It can be seen that in both approaches the composition of the portfolio is the same and the weights are very close. The reason for this similarity is that our portfolio is not too big and we have enough data points compared to the number of the securities.

It's time to evaluate the performance of the portfolio during the testing period. To understand the situation better we are going to compare the performance for both train and test periods.

Portfolio stats:

| | SampleΣ | | ShrinkΣ | |
|---|---|---|---|---|
| | Train | Test | Train | Test |
| μ_ann | 0.074 | 0.181 | 0.071 | 0.185 |
| σ_ann | 0.130 | 0.176 | 0.129 | 0.175 |
| Sharpe | 0.572 | 1.029 | 0.553 | 1.059 |
| TotalReturn | 0.469 | 0.070 | 0.451 | 0.072 |

As expected both portfolios --constructed using different approaches to estimating the covariance-- have a very similar performance. There some points to discuss here

- The Shario ratio in the training process is less than one. It usually is interpreted that the portfolio composition is not good and the excess risk is not compensated by enough return.
- The volatility in the testing period is substantially higher than the training period. It shows that the market is passing through an unstable high fluctuations phase.
- The total return and also Sharpe ratio are negative for the evaluation (test) period. So, we are losing money!
- Looking at the news we can see why we are facing this situation. In our interpretation, the global financial markets, not only the US, is experiencing a high volatility time due to new tariff policies of the US government.

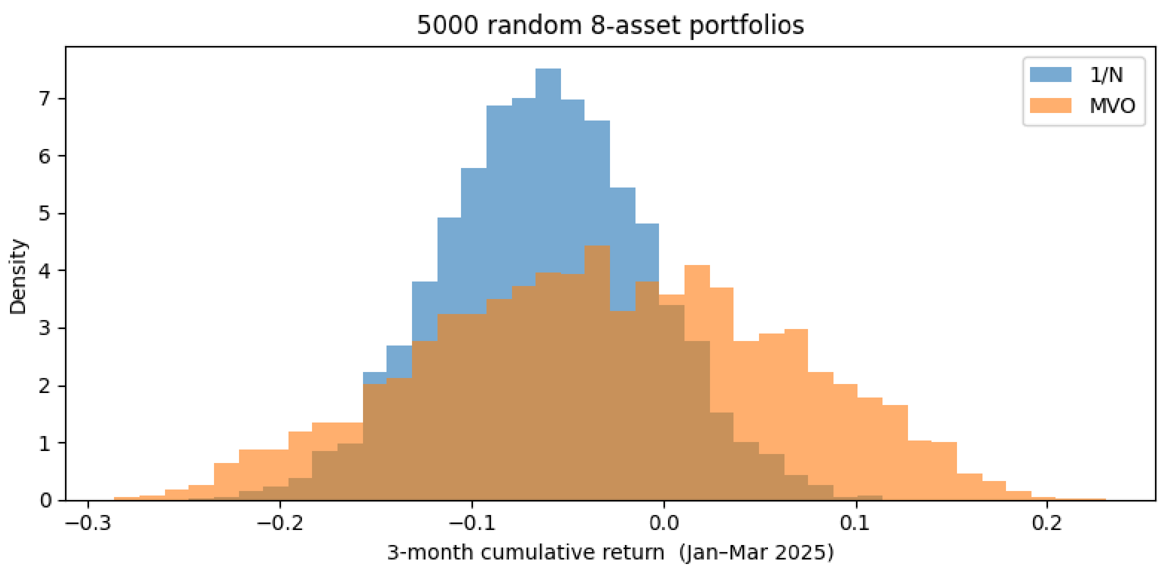
Risk contribution table (train Σ):

| Asset | SampleΣ | ShrinkΣ |
|---|---|---|
| **GC=F** | 9.0% | 9.5% |
| **KMB** | 27.6% | 28.7% |
| **MDT** | 10.8% | 11.1% |
| **NVDA** | 27.5% | 25.1% |
| **SNPS** | 13.8% | 14.2% |
| **V** | 11.3% | 11.5% |

We can see the highest risk contribution by KMB, NVDA followed by V and SNPS.

## Step 3: Random Strategy Optimization

To see if we could have a better performance, let's compare two situations. The strategy is as follows. We randomly choose 8 assets from our universe and construct an equal-weights (1/N) portfolio. Then we compare the performance of this portfolio with a portfolio in which its weights are calibrated by mean-variance optimization (MVO). The performance evaluation is done for the testing period. We did this process 5000 times and the result is displayed in the below graph. We used MonteCarlo simulation to allocate equally across 8 randomly selected portfolios iteratively and then calculate cumulative returns.



From the plot, it seems that the portfolios based on the MVO performed better than equally-weighted ones.

## Step 4: Black-Litterman

First, we must declare our views as the portfolio manager to obtain the prior values to feed into the Black-Litterman (BL) model. We did a search into the technical news and the report on the dates before 2024/12/31 to gather relevant news about the firms in our universe.

## Views

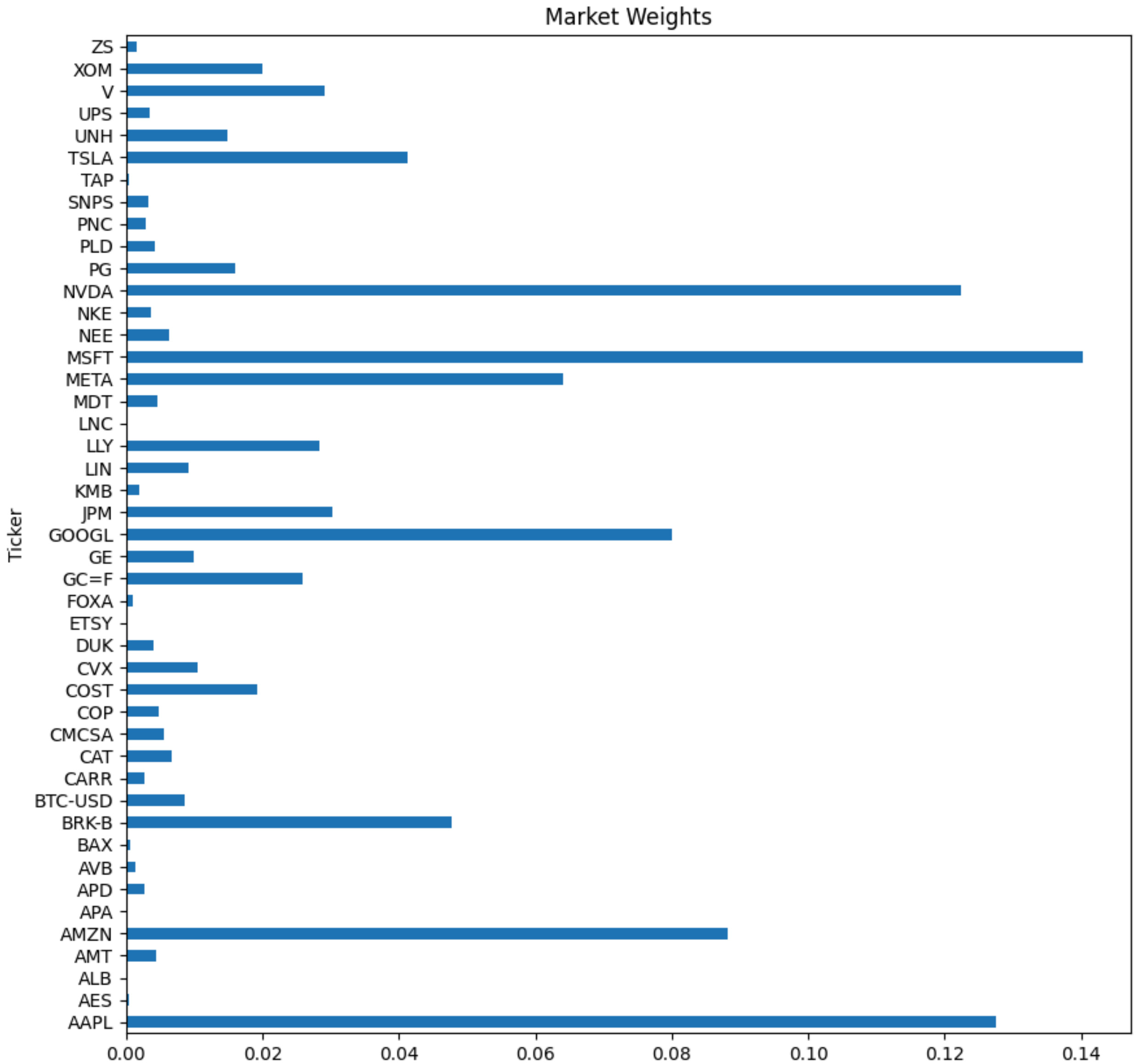
**Material Sector**

- Linde (LIN) – Absolute view. Q: ~+10%. Confidence: Medium. Justification: The company underperformed in 2024 but it is expected to improve in 2025, if the global economy improves. CNBC (Dec 27, 2024) noted Linde could return to its “beat-and-raise” earnings cadence with a stronger economy. The company ended Q3 2024 with a high record of backlos that is a positive factor for growth. [https://www.cnbc.com/2024/12/27/a-pickup-in-the-global-economy-should-get-linde-back-to-its-winning-ways-in-2025.html]
- Air Products & Chemicals (APD) – Absolute view. Q: ~+15%. Confidence: Medium. Justification: An activist investor called Mantle Ridge took more than $1B stake in APD in 2024, pushing for strategic changes (succession planning, capital allocation) to unlock value. The stock increased about 9% on the news in Oct 2024. With these catalysts, analysts expect improved efficiency and shareholder returns. [https://www.investopedia.com/air-products-stock-jumps-as-activist-investor-pushes-strategy-shift-8724566]
- Albemarle (ALB) – Absolute view. Q: ~+15%. Confidence: Medium (high upside, but commodity-driven). Justification: ALB’s shares were depressed in 2024 due to lower global lithium prices, but analysts anticipate a strong recovery due to the increasing demand for batteries. The average analyst target implies ~15-20% rise from late-2024 prices. [https://www.zacks.com/stock/quote/ALB]

**For other Views on Companies/Sector: <pls refer jupiter or pdf of jupiter>**

The above views can be used to construct the P, Q and Ω matrix that reflects the opinions of the portfolio manager.

One of the main ingredients of the BL model is the market weight of each asset in our universe. We used the current market cap of the firms in the S&P 500 as a proxy for the weights on the final of 2024. For the market cap of Gold and Bitcoin we used some big but somehow arbitrary values. The result is as follow




Using this market cap and the δ we arrived at the below priors
<Pls refer jupiter or pdf of jupiter>

Because Σ w is the covariance of asset i with the cap-weighted market portfolio, the sign depends on the asset's beta:

- Positive beta -> positive Π (asset expected to earn a positive excess return over the risk-free rate).

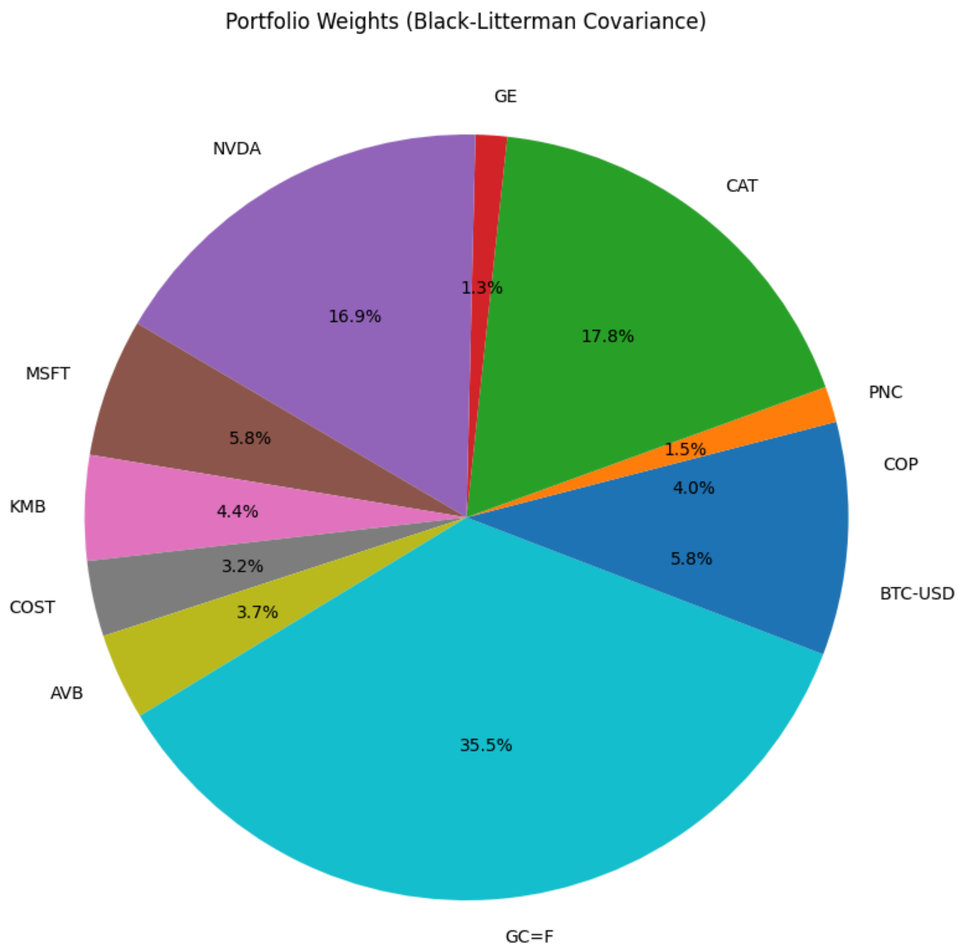
- Negative or very low beta -> zero or negative Π (CAPM says it should earn less than the risk-free rate because it hedges the market).

So negative securities are not an error; they simply indicate assets that reduce portfolio volatility (low/negative beta) and therefore carry a lower equilibrium reward.

Let's evaluate the performance of the portfolio constructed by means of Black-Litterman model and compare it with the previous ones. Before the comparison we prone the weights less than 1% in the BL model to have better visualizations.

| | MVO_Σsample | MVO_Σshrink | BL |
|---|---|---|---|
| COP | 0.0 | 0.0 | 0.0402 |
| PNC | 0.0 | 0.0 | 0.0151 |
| V | 0.107 | 0.109 | 0.0 |
| CAT | 0.0 | 0.0 | 0.1776 |
| GE | 0.0 | 0.0 | 0.0133 |
| NVDA | 0.103 | 0.097 | 0.1691 |
| SNPS | 0.079 | 0.082 | 0.0 |
| MSFT | 0.0 | 0.0 | 0.0584 |
| KMB | 0.347 | 0.346 | 0.0444 |
| COST | 0.0 | 0.0 | 0.0319 |
| MDT | 0.122 | 0.124 | 0.0 |
| AVB | 0.0 | 0.0 | 0.0366 |
| GC=F | 0.243 | 0.242 | 0.3551 |
| BTC-USD | 0.0 | 0.0 | 0.0582 |

BL has given max weights to GC=F followed by CAT, NVDA,BTC-USD, MSFT, KMB whereas MVO has given max weights to KMB, followed by GC=F, NVDA,V and MDT. The weights of composing assets in the portfolio obtained by the BL model are




Contribution to portfolio variance (train Σ): (below)

| | MarkovΣshrink | BL |
|---|---|---|
| AVB | 0.0 | 1.9425 |
| BTC-USD | 0.0 | 11.2078 |
| CAT | 0.0 | 18.8715 |
| COP | 0.0 | 3.0573 |
| COST | 0.0 | 2.1356 |
| GC=F | 9.4533 | 12.8767 |
| GE | 0.0 | 1.1947 |
| KMB | 28.6554 | 0.4447 |
| MDT | 11.0837 | 0.0 |
| MSFT | 0.0 | 5.7416 |
| NVDA | 25.1443 | 41.3959 |
| PNC | 0.0 | 1.1316 |
| SNPS | 14.1591 | 0.0 |
| V | 11.5042 | 0.0 |

Highest contribution to variance by MVO is by KMB, NVDA, SNPS, V.MDT and GC=F.
And finally the most important part, the performance of the models.
Highest contribution to variance for BL model is by NVDA, followed by CAT, BTC-USD and GC=F.
Risk-return tradeoff: Well,it is said that for higher returns one need to take higher risks..

```
=== In-sample (2021-24) ===
```

| | MarkovΣshrink | BL |
|---|---|---|
| μ_ann | 0.071 | 0.185 |
| σ_ann | 0.129 | 0.169 |
| Sharpe | 0.553 | 1.094 |

```
=== Out-of-sample (Jan–Mar 2025)
```

| | MarkovΣshrink | BL |
|---|---|---|
| μ_ann | 0.185 | 0.033 |
| σ_ann | 0.175 | 0.242 |
| Sharpe | 1.059 | 0.135 |

The performance of our portfolio is not as expected. In fact, since the beginning of the year 2025 due to the tariff policies of the US government, the financial market has been very volatile and unstable. Somehow, we are experiencing a regime shift in the market. So, the importance of the historical data on the future performance of the portfolio is reduced.

## 4.3 Background on Black Litterman model and implementation

The Black-Litterman (BL) model is an Bayesian asset allocation framework for a portfolio that combines market equilibrium returns with investor specific views. It was introduced in 1992 to overcome limitations of traditional mean-variance optimization (MVO), such as extreme and unstable weights due to input sensitivity [1].

**1. Market Equilibrium Prior: The Implied Returns**

The implementation of BL model starts with getting the market-implied equilibrium returns, derived rom the Capital Asset Pricing Model (CAPM):

$\pi = \delta \cdot \Sigma \cdot wm$

$\delta = E[Rm] - rf\sigma m\hat{}2$

where:

- pi: Implied excess returns (prior)
- δ: Risk aversion coefficient, estimated from historical market data
- Σ: Covariance matrix of returns (we used a shrinkage estimator)
- wm: Market capitalization weights (current market portfolio)

## 2. Absolute views and Relative views as Investor Views

Investor views are encoded using a set of equations that expresses beliefs about future asset returns:

$P\mu = Q + epsilon$

where:

- $P \in Rk \times n$: The P matrix representing how each view relates to the assets.
- $Q \in Rk$: Expected return of each view (can be absolute or relative).
- epsilon: A normally distributed error term with covariance Ω (Omega) ~N(0,Ω): represents the error or uncertainty in each view

Absolute views correspond to investors beliefs about an asset's expected return (e.g., E[r_i] = q or Asset A will return 5%), while relative views express expectations about differences in performance (e.g., E[ri−rj] = q or Asset A will outperform Asset B by 2%).).

.The matrix Ω (Omega) plays an important role, as it determines how much influence each view has on the final posterior. A more confident view (smaller uncertainty) will shift the posterior returns closer to the view.

$\Omega = diag(\tau P \Sigma P^{\wedge T} / \text{ confidence\_scale}^{\wedge 2})$

where:

- τ (tau): A small scalar (often between 0.025 and 0.05) reflecting the uncertainty in the prior.
- $P\Sigma P^{\wedge T}$: The variance of the portfolio under the prior covariance.
- confidence_scale: A scalar from a predefined scale: e.g.

  High confidence: small denominator (e.g., 0.1)

Medium confidence: moderate (e.g., 0.3)

Low confidence: higher uncertainty (e.g., 1.0)

This formulation will ensure that each view's uncertainty is properly adjusted for both its internal volatility and the investor's stated confidence level

### 3. Posterior Distribution

The BL model updates the prior mean π (pi) and covariance Σ (Sigma) using Bayesian inference to obtain the posterior estimates:

$M\_BL=[(\tau\Sigma)^{\wedge-1}+P^{\wedge}T\Omega^{\wedge-1}P]^{\wedge-1}[(\tau\Sigma)^{\wedge-1}\pi+P^{\wedge}T\Omega^{\wedge-1}Q]$

$\Sigma\_BL=\Sigma+[(\tau\Sigma)^{\wedge}-1+P^{\wedge}T\Omega^{\wedge}-1P]^{\wedge}-1$

These adjusted posterior moments reflect both market equilibrium and investor views.

### 4. Optimization and Post-processing

The posterior μ_BL and Σ_BL are given as inputs to our MVO weights function :

$Max\_w( w^{\wedge}T\mu\_BL-\lambda/2w^{\wedge}T\Sigma\_BLw$ subjected to: $w\in[w\_min,w\_max]$, and $\sum w=1$

After solving for the optimal weights, we apply a pruning function:

prune_weights(w, floor=0.01)

- This eliminates negligible weights (less than 1%) to ensure the final portfolio is implementable and sparse

### 5. Interpretation

- **π** (pi) represents the best guess from the market without any subjective inputs.
- Views allow overlaying investor beliefs in a structured, quantitative way.
- Ω (Omega) captures uncertainty in views, allowing flexible expression of confidence.
- Posterior returns and covariance reflect a blend of market equilibrium and adjusted beliefs.

### 6. Implementation Notes

- We have used shrinkage estimators for Σ (Sigma) to improve numerical stability.

- Market weights are obtained using data from yahoo finance) and adjusted manually for non-equity assets like gold and Bitcoin.
- The final output compares the BL weights with traditional MVO solutions.
- Below Inputs and outputs table:

| Element | Symbol | Description |
|---|---|---|
| Prior mean | Π (pi) | Implied returns from CAPM |
| Prior cov. | Σ (Sigma) | Shrinkage estimator |
| View matrix | P | Which assets each view targets |
| View returns | Q | Expected returns from views |
| View variance | Ω (Omega) | Uncertainty of views |
| Posterior mu | μ_BL | Adjusted expected returns |
| Posterior cov | Σ_BL | Adjusted covariance matrix |

## Step 5: Kelly Criterion

### 1. The Kelly Formula

For asset with expected return $\mu$, variance $\sigma^2$, and risk-free rate $r_f$, the Kelly fraction $f^*$ is given by [2]:

$$f^* = (\mu - r_f) / \sigma^2$$

where:
$\mu$: Expected return of the asset (usually daily or annualized)
r_f: Risk-free rate (e.g., daily treasury yield)
$\sigma^2$: Variance of the asset's returns
f*: Optimal fraction of portfolio to allocate to that asset

This maximizes expected logarithmic growth of wealth:

max_f E[log(1 + fR)]

**2. For Multi Asset Portfolios**

In a portfolio context, the Kelly rule becomes:

$w^* = \Sigma^{-1} (\mu - r\_f)$

where:
$\mu$: Vector of expected excess returns
$\Sigma$: Covariance matrix of returns
w: Portfolio weights

**3. Inputs**

We have already expected return| $\mu$. historical mean returns (daily), Risk-free rate, Covariance, $\Sigma$, Sample or shrinkage estimator |

**4. Backtesting Procedure**

1. Estimate kelly weights per asset using in-sample returns_train data.
2. Apply weights to test data and calculate portfolio growth i.e. cumulative returns:
   Port_val_t = Prt_val_{t-1} (1 + w^T r_t)
3. Compare kelly returns against:
   Equal-weight (1/N)
   Mean-variance (MVO)
   Black-Litterman (BL)
4. Evaluate final wealth, Sharpe ratio, volatility.

**5. Interpretation and Future Considerations**

Full Kelly is aggressive; fractional Kelly is common in practice.
Results are sensitive to input estimation.

Kelly adjusts for diversification and covariance like other MVO and BL models.
Weekly/monthly rebalancing is recommended for stability.

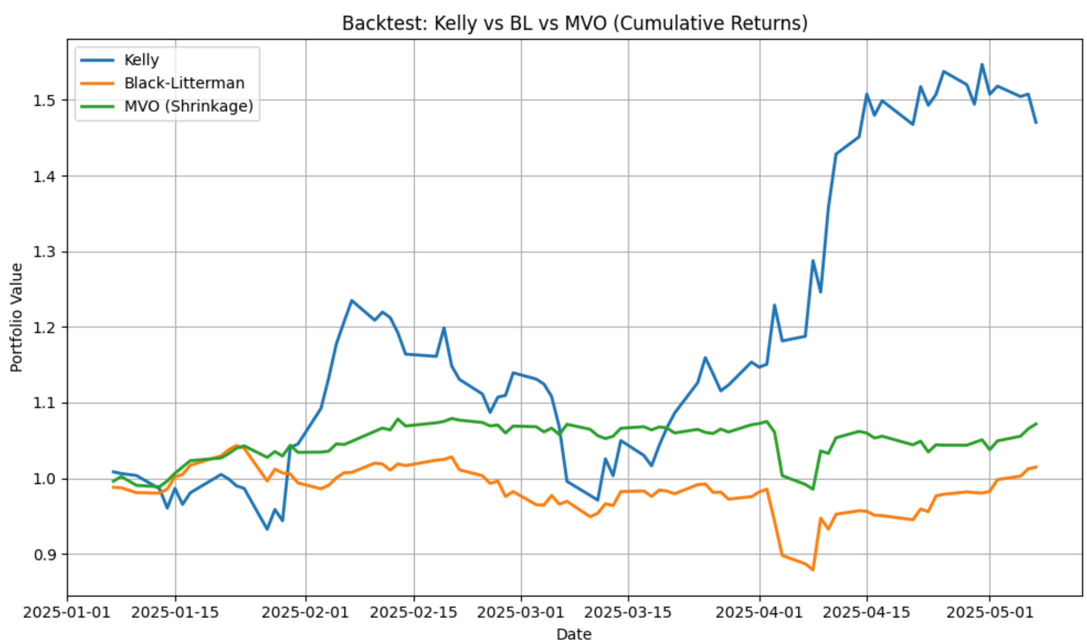
**6. Kelly Results and Comparison**

Kelly offers optimal growth when calibrated correctly.

Kelly Performance:

```
Ann. Return        1.2908
Ann. Volatility    0.4927
Sharpe Ratio       2.5576
Total Return       0.4703
dtype: float64
```

Better Sharpe Ratio and better growth than MVO and BL



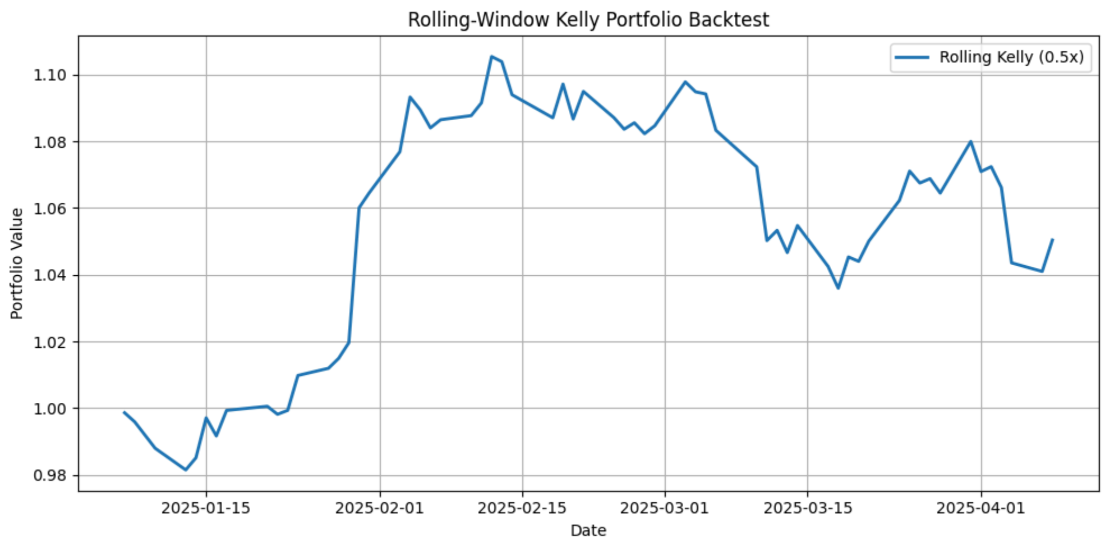

**7. Rolling-Window Kelly Backtest (with rebalancing)**

We can use fractional Kelly (e.g., 0.5x) for reduced volatility
Rolling-window estimation for dynamic rebalancing

**Steps:**

- Re-estimate fractional (0.5) Kelly weights every N days (e.g., 21)

- Applies weights to the next N days
- Accumulates portfolio value over time




**Final Notes:**

◦ BL: combines market belief and investor views

◦ We compared performance to MVO and Kelly under test conditions.
◦ BL has given max weights to GC=F followed by CAT, NVDA,BTC-USD, MSFT, KMB whereas MVO has given max weights to KMB, followed by GC=F, NVDA,V and MDT.

◦ Highest contribution to variance for BL model is by NVDA, followed by CAT, BTC-USD and GC=F.

◦ Risk-return tradeoff: Well, it is said for higher returns one need to take higher risks..
◦ Highest contribution to variance for MVO is by KMB, NVDA, SNPS, V.MDT and GC=F

◦ In long run Kelly outperformed MVO and BL because of many assets going short and ability to fractional kelly and monthly rebalancing

◦ High volatility do impact the prediction ability

# References:

[1] Meucci, Attilio. "The Black-Litterman Approach: Original Model and Extensions." *SSRN Electronic Journal*, 1 Aug. 2008, https://doi.org/10.2139/ssrn.1117574.

[2] Carta, Andrea, and Claudio Conversano. "Practical Implementation of the Kelly Criterion: Optimal Growth Rate, Number of Trades, and Rebalancing Frequency for Equity Portfolios." *Frontiers in Applied Mathematics and Statistics*, vol. 6, 2020, article 577050. https://doi.org/10.3389/fams.2020.577050.

**Communication Services**

- -Meta Platforms (META) – Absolute view. Q: ~+15%. Confidence: High. (https://investorsobserver.com/news/meta-platforms-meta-stock-prediction-for-2025-betting-on-ai)
- Comcast (CMCSA) – Absolute view. Q: ~+5%. Confidence: Medium.. (https://www.nasdaq.com/articles/comcasts-q4-2024-earnings-what-expect)
- Fox Corp (FOXA) – Absolute view. Q: ~+5%. Confidence: Medium.. (https://www.nasdaq.com/articles/what-expect-fox-corporations-q2-2025-earnings-report)
- Alphabet/Google (GOOGL) – Absolute view. Q: ~+10%. Confidence: High. (https://www.investors.com/news/technology/google-stock-buy-now-alphabet-stock-december-2024/)
- Relative view – AAPL vs. MSFT: Microsoft is expected to outperform Apple. Q: +~7% (MSFT over AAPL). Confidence: High. (https://www.nasdaq.com/articles/will-microsoft-still-be-worth-more-than-apple-by-2025, https://schwabnetwork.com/video/2024-year-in-review-a-i-s-role-in-aapl-msft-nvda-)

**Energy Sector**

- Exxon Mobil (XOM) – Absolute view. Q: ~+15%. Confidence: High. (https://www.nasdaq.com/articles/3-reasons-why-exxon-xom-set-excellence-2025)
- ConocoPhillips (COP) – Absolute view. Q: ~-10%. Confidence: Medium. (https://seekingalpha.com/article/4746461-conocophillips-attractive-even-in-a-more-muted-commodity-price-environment)
- APA Corp (APA) – Absolute view. Q: ~-5%. Confidence: Medium-Low. (https://finance.yahoo.com/news/apa-third-quarter-2024-earnings-121345607.html)
- Chevron (CVX) – Absolute view. Q: ~-5%. Confidence: Medium. (https://www.nasdaq.com/articles/chevrons-spending-shift-should-you-buy-stock-2025)

**Financials**

- JPMorgan Chase (JPM) – Absolute view. Q: ~+7%. Confidence: Medium. (https://www.globenewswire.com/news-release/2024/11/18/2982958/28124/en/2024-J-P-Morgan-Private-Bank-Competitor-Profile-Key-Strategies-Financial-Performance-Customers-and-Products-Brand-Building-Activities.html, https://www.investopedia.com/what-analysts-think-of-jpmorgan-chase-stock-ahead-of-earnings-8771652)
- PNC Financial (PNC) – Absolute view. Q: ~+7%. Confidence: Medium. (https://www.investopedia.com/what-analysts-think-of-jpmorgan-chase-stock-ahead-of-earnings-8771652)
- Lincoln National (LNC) – Absolute view. Q: ~+5%. Confidence: Medium-Low. (https://finance.yahoo.com/news/lincoln-national-q3-earnings-beat-184700902.html, https://www.businesswire.com/news/home/20241101089053/en/Lincoln-National-Corporations-Board-of-Directors-Declares-Series-D-Preferred-Stock-Dividend)
- Berkshire Hathaway (BRK.B) - Absolute view. Q: ~+10%. Confidence: High. (https://www.fool.com/investing/2024/12/08/cash-and-patience-berkshires-2024-moves-set-the/)
- Visa (V) - Absolute view. Q: +10%. Confidence: High. revenue. (https://finance.yahoo.com/news/visa-full-2024-earnings-line-130723645.html)